

Document Control

- Document Title: Business Requirements Document (BRD)
- System Name: Application
- Version: 1.0
- Date: Not present in source JSON
- Author: Not present in source JSON
- Status: Draft

Executive Summary

- This BRD outlines the ingestion, cleansing, transformation, and aggregation of customer and order CSV data into Delta tables using Delta Live Tables in the Application system.
- Focus is on SCD Type 2 handling for customer changes and creation of an aggregate spend table with specified lineage and rules.

Business Objectives

- Load source CSVs into Delta tables for customer and order data.
- Cleanse data by removing "Null"/Null literals and duplicates.
- Enrich orders with computed TotalAmount.
- Join customer and order to produce an ordersummary table with SCD Type 2 change handling.
- Produce daily customer spend aggregates in customeraggregatespend.
- Implement the pipeline using Delta Live Tables.

In-Scope

- Source ingestion from file system.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Data cleansing as specified.
- Computation of TotalAmount field.
- Join logic and SCD Type 2 handling for customer changes in ordersummary.
- Aggregate processing for customeraggregatespend.
- Delta Live Tables implementation.

Out-of-Scope

- Systems or fields not in source JSON.
- Security, access controls, SLAs, monitoring, alerting.
- Performance and scaling parameters.
- Scheduling and orchestration.

- Data quality metrics beyond cleansing actions.
- User interface, reports, analytics consumption.

Stakeholders

- Not present in source JSON.

Business Requirements

BR-BI-001 — Source Ingestion

- Description: Load customer CSV and order CSV into respective Delta tables.
- Business Value: Enables reliable raw data intake for downstream processing.
- Priority: High

BR-BI-002 — Schema Definition

- Description: Customer table with CustId, Name, EmailId, Region; Order table with OrderId, ItemName, PricePerUnit, Qty, Date, CustId.
- Business Value: Ensures consistent entity modelling and downstream join compatibility.
- Priority: High

BR-DQ-003 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from customer and order tables.
- Business Value: Improves data quality and accuracy of analytics.
- Priority: High

BR-EN-004 — Computed Column TotalAmount

- Description: Add TotalAmount column to order calculated as PricePerUnit × Qty.
- Business Value: Supports spend analytics and aggregation.
- Priority: High

BR-TR-005 — Ordersummary Table Creation

- Description: Create ordersummary Delta table as joined output of customer and order.
- Business Value: Centralizes customer order facts for analytics and historic tracking.
- Priority: High

BR-TR-006 — SCD Type 2 Handling in Ordersummary

- Description: Implement SCD Type 2 logic for customer changes; maintain Active/Inactive, StartDate, EndDate fields.
- Business Value: Enables full history and auditability for customer dimension.

- Priority: High

BR-TR-007 — Aggregate Table Creation

- Description: Create customeraggregatespend by aggregating sum(TotalAmount) grouped by Name and Date.
- Business Value: Provides actionable KPIs for daily spend reporting.
- Priority: High

BR-IM-008 — End-to-End Delta Live Tables Pipeline

- Description: Implement all pipeline steps using Delta Live Tables with defined dependencies.
- Business Value: Streamlines operationalization and repeatability.
- Priority: High

Assumptions & Constraints

- No assumptions made beyond explicit requirements.
- Delta Live Tables is the prescribed platform.
- StartDate, EndDate, Active assumed for SCD Type 2 but explicit types/definitions are not specified.
- Any missing technical fields or parameters are not present in source JSON.

Success Metrics

- 100% of customer and order source data correctly loaded into Delta tables.
- Deduplication and null literal removal demonstrated in target data.
- All computed and aggregate fields accurately populated.
- SCD Type 2 changes reflected in historical customer attributes.
- Daily spend totals match business logic for all processed data.
- Pipeline executes end-to-end in Delta Live Tables without manual intervention.

Risks & Mitigations

- Potential ambiguity in SCD Type 2 tracking field types or names: Flag for confirmation with data architect.
- Data quality limited to cleansing rules provided: Recommend future enhancement of profiling routines.
- Error handling and monitoring requirements unspecified: Review implementation post-pilot, add as needed.
- Performance scaling not addressed: Assess manually during initial load; revisit for large volumes.

Approval

- Business Owner: Not present in source JSON

- Data Architect: Not present in source JSON
- Date Approved: Not present in source JSON

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Load as Delta table 'customer'
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Load as Delta table 'order'

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Bronze	Delta	CustId, Name, EmailId, Region	Ingest from customer CSV
	Bronze	Delta	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Ingest from order CSV
y	Silver	Delta	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	Join of customer and order
egatespend	Gold	Delta	Name, TotalAmount, Date	Aggregation by Name, Date

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Remove "Null"/Null literals	customer, order	customer, order	Filter out any 'Null' strings/nulls
2	Deduplicate	customer, order	customer, order	Remove duplicate records
3	Compute TotalAmount	order	order	PricePerUnit × Qty
4	Join	customer, order	ordersummary	Join on CustId
5	SCD Type 2	ordersummary	ordersummary	Set Active/Inactive; update StartDate/EndDate
6	Aggregate	ordersummary	customeraggregatespend	Group by Name, Date; sum on TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Computation	TotalAmount = PricePerUnit × Qty	Provided requirement
BR-2	Cleansing	Remove "Null"/Null, deduplicate	Provided requirement
BR-3	SCD Type 2	Maintain full customer attribute history	Provided requirement
BR-4	Aggregation	Daily spend per Name	Provided requirement

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customer and order CSVs	Source paths

2	Cleanse nulls and duplicates	Step 1
3	Compute TotalAmount in order	Step 2
4	Join to create ordersummary table	Step 3
5	Apply SCD Type 2 to customer in ordersummary	Step 4
6	Aggregate to customeraggregatespend	Step 5

Appendix F: Data Flow Diagram

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